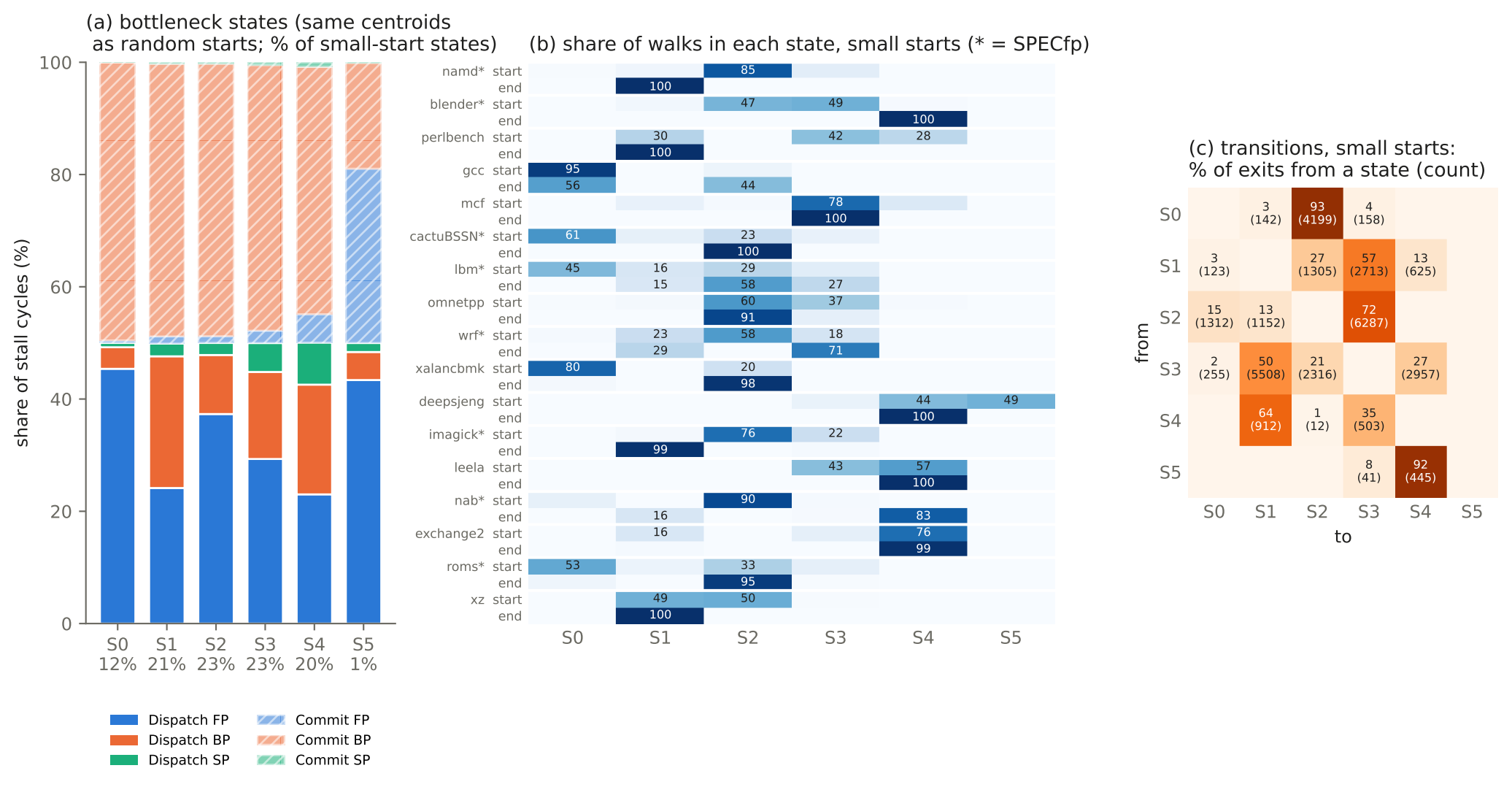
(a) bottleneck states (same centroids as random starts; % of small-start states)
share of stall cycles (%)
100
80
60
40
20
0
S0 12%
S1 21%
S2 23%
S3 23%
S4 20%
S5 1%
Dispatch FP
Dispatch BP
Dispatch SP
Commit FP
Commit BP
Commit SP
(b) share of walks in each state, small starts (* = SPECfp)
namd* start 85
end 100
blender* start 47 49
end 100
perlbench start 30 42 28
end 100
gcc start 95
end 56 44
mcf start 78
end 100
cactuBSSN* start 61 23
end 100
lbm* start 45 16 29
end 15 58 27
omnetpp start 60 37
end 91
wrf* start 23 58 18
end 29 71
xalancbmk start 80 20
end 98
deepsjeng start 44 49
end 100
imagick* start 76 22
end 99
leela start 43 57
end 100
nab* start 90
end 16 83
exchange2 start 16 76
end 99
roms* start 53 33
end 95
xz start 49 50
end 100
S0 S1 S2 S3 S4 S5
(c) transitions, small starts:
% of exits from a state (count)
from
S0 3 (142) 93 (4199) 4 (158)
S1 3 (123) 27 (1305) 57 (2713) 13 (625)
S2 15 (1312) 13 (1152) 72 (6287)
S3 2 (255) 50 (5508) 21 (2316) 27 (2957)
S4 64 (912) 1 (12) 35 (503)
S5 8 (41) 92 (445)
S0 S1 S2 S3 S4 S5
to